

MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Computer Science and Engineering

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<<Software Project for Course Software Engineering>>

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Chapter 1

Introduction

1.1 Purpose

This document is the User Requirements Specification (URS) for **TrajVis Studio**, the system proposed for Group 1 in Task 1. It states what the system must do, from the point of view of the people who will use it and of the people who will pay for it.

The document serves the build team, two surveyed user representatives and the course assessors. The assigned customer (the last group on the Part A roster) could not be reached after repeated contact; individual members of two other groups in the class were therefore surveyed as user representatives. Appendix A records this limitation and the traceability from each requirement to its source.

This specification builds on, and does not replace, the project proposal submitted as Task 1. Where the proposal said what the team intends to deliver, this document states the requirements in a form that can be tested one by one.

1.2 Notation

This specification and its supporting diagrams use the Unified Modelling Language (UML) notation, as the template requires. In addition:

- Every functional requirement carries an identifier `REQ-x.y` and every non-functional requirement an identifier of the form `PERF-x.y`, `SEC-x.y`, `AVA-x.y` or `CAP-x.y`.
- The word *shall* marks a binding requirement; *should* marks a preference that may be traded against cost.
- Source identifiers `S0`–`S5`, used in Appendix A, name where each requirement came from; *shall* in a table cell is binding in the same way as in the running text.

1.3 Scope

1.3.1 In scope

The system accepts a question about recorded movement, written in ordinary English or Chinese, and returns a finished figure with a plain-language summary, together with a record that can be re-run. It covers the whole round trip: understanding the question, retrieving and filtering the records, choosing and drawing the chart, describing the result, and measuring its own accuracy.

1.3.2 Out of scope

The system does not collect the movement data, does not clean it, and does not act as a geographic information system. It does not make planning decisions; it answers questions about records that already exist. It offers no mobile application and no public web service.

1.4 Context Diagram

Figure 1-1 shows the boundary of the system: what lies inside it, which outside parties exchange information with it, and what is exchanged.

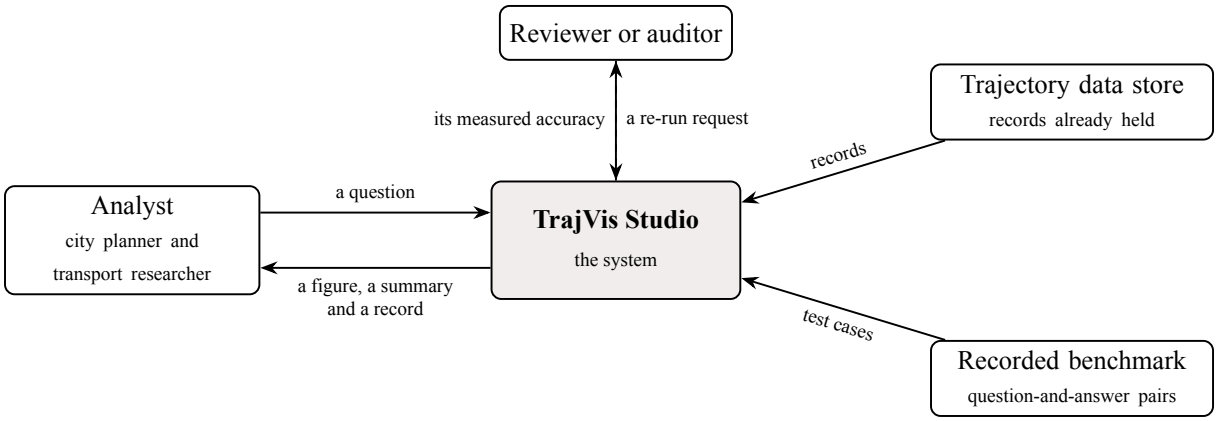


Figure 1-1: Context diagram: the system and the parties it exchanges information with.

1.5 Definitions and Acronyms

Term	Meaning in this document
Trajectory	The recorded path of one moving object over time, as a sequence of timestamped positions.
Analyst	A person who needs an answer about recorded movement but cannot write a data query.
Exchange	One question put to the system together with everything the system returned for it.
Record	The stored form of an exchange: the question, the request the system derived, the data snapshot identifier and the figure.
Benchmark	A published set of recorded question-and-answer pairs used to measure the accuracy of the automatic interpretation.
Snapshot	Immutable identifier of the data set an exchange was answered from, so that the answer can be reproduced later.
Chart specification	The chart type, fields, filters, scale and time zone chosen for a figure.
Confidence notice	The visible statement of how certain the system is that it read the question correctly; at low confidence no single reading is chosen without the user's confirmation.
URS	User Requirements Specification, this document.

1.6 References

1. The team's Task 1 submission, *TrajVis Studio: A Self-Service Platform for Natural-Language Trajectory Analysis*, Group 1, 2026.
2. Project repository: https://github.com/chenhongzhou666/MUST-SE-TrajVis
3. Task 2 description, *User Requirements Specifications*, Moodle, 2026.09.29.
4. The template supplied with Task 2, *Task2-User Requirements Specifications Template*, Moodle, 2026.09.29.

The six writing guides listed in Part D of the task description were read while preparing this document; they are not cited individually.

1.7 Overview

Chapter 2 describes the system and its users in general terms. Chapter 3 states the software and hardware constraints and Chapter 4 the assumptions and dependencies. Chapter 5 lists every functional requirement in summary form and Chapters 6–11 define each one in full. Chapter 12 covers the external interfaces and Chapter 13 the non-functional requirements. Chapter 14 records the requirements that come from the domain rather than from a user. Appendix A records every elicitation source and traces each requirement back to one.

1.8 Contributions

Chen Hongzhou wrote this specification. Zhou Guoxi is the team's contact with the assigned customer; Xu Jing prepared the questionnaire summarised in §A.3 and collected the responses that are cited as `S1` and `S2`. The work after this document will be divided among the three members along the lines set out in the Task 1 proposal.

Chapter 2

General Description

2.1 System Functions

The system answers questions about recorded movement. A person asks a question in ordinary language—*which streets stayed busy after nine in the evening last week*—and receives a figure that answers it, a sentence saying what the figure shows, and a record of how the answer was produced.

Its purpose is to remove the queue between the person who has the question and the person who can query the data. That round trip takes days today and leaves a picture nobody can check afterwards; the system closes it within minutes and keeps a record.

At the highest level the system: turns the question into an exact written request; finds and filters the records the request refers to; chooses and draws the chart that fits the question; describes the result in one or two sentences; and measures its own accuracy against a recorded benchmark on demand.

The functions are enumerated as requirements in Chapter 5. *(Source: Task 1, §3.4)*

2.2 User Characteristics

Table 2-1: User roles and their characteristics.

Role	What they need	Interface type	Platform
Analyst	An answer to a question about movement, without writing a query	Self-service, untrained; must work without a manual	Desktop web browser
Reviewer or auditor	To see how a past answer was produced and to re-run it	Trained user; occasional	Desktop web browser
Stakeholder	Review the requirements and accept delivery.	Documents and review meetings.	No runtime role; reads reports only.

No role requires installation, a licence key or an account on a third-party service. Questionnaire respondents used desktop computers (macOS and Windows); no mobile use was requested.

A ByteDance researcher who works on movement and footprint data has described this kind of question as a live requirement there; the source is `S5` in Appendix A, recorded as informal because it came without a document and supports nothing on its own.

Chapter 3

General Constraints

3.1 Software Constraints

- The system shall be written in Python, because the published method it builds on is distributed in Python and re-implementing it in another language would discard the working prototype that makes the project feasible. *(Source: Task 1, §3.8)*
- The system shall run as a web application and be usable from a current desktop browser with no client-side installation.
- The system shall reuse existing, openly licensed components rather than re-implement them. The parts already identified, with their cost and readiness, are listed in Table 3-1. *(Source: Task 1, Table 3-3)*
- The system shall not require any commercial software licence or paid cloud service.

Table 3-1: Parts to be reused, with cost and readiness.

Part	Licence	Cost	Readiness
Published interpretation method	Open	None	Already reproduced by the team, reaching 64% fully-correct
Public trajectory benchmark	Open	None	9,608 recorded question-and-answer pairs, downloaded and in use
Chart rendering library	Open	None	Mature; no work required beyond calling it
Map and place-name data	Open	None	In use for resolving place names written in a question

Open-source components only; each licence name is recorded in the project repository.

3.2 Hardware Constraints

- The system shall run on an ordinary 24 GB laptop for development and single-user demonstration; no server or cloud resource is assumed for that environment.
- Production multi-user use requires a supported server or workstation.
- Users shall need no hardware beyond a desktop or laptop computer with a pointing device; respondents used macOS and Windows machines and asked for nothing more.

Chapter 4

Assumptions and Dependencies

Table 4-1: Assumptions and dependencies, with the consequence if each fails.

Id	Assumption or dependency	Consequence if it does not hold
A1	The recorded movement data is already held by the customer in a usable form.	The project would need a data-preparation stage that is not in the plan.
A2	The published benchmark may be used freely and remains available at the stated address.	Accuracy could not be measured by the system itself and M3 could not be verified.
A3	Analysts are willing to ask questions in ordinary language rather than in a query language.	The central feature of the system would be unused.
A4	Trajectory source records are de-identified; free-text questions may still contain personal data.	A redaction and access-control stage would be required before delivery.
A5	The team has continued access to the machines and the network used for development.	Development would stop; the plan assumes continuous access.

Chapter 5

Functional Requirements Master List

This chapter lists the functional requirements in summary form. Each is defined in full in the chapter named in the last column.

Table 5-1: Functional requirements master list.

Req. ID	Requirement Name	Requirement Description	Defined in
REQ-1.1	Ask in ordinary language	The user states a question in plain English or Chinese and the system turns it into an exact written request.	Ch 6
REQ-1.2	Automatic retrieval and filtering	The system finds the trajectories the request refers to, resolves place names against the records, and returns only the matching subset.	Ch 7
REQ-1.3	Automatic chart selection and drawing	The system chooses the chart that fits the question and renders the figure, with no chart type or plotting tool chosen by the user.	Ch 8
REQ-1.4	Plain-language summary of the result	The system describes what the figure shows in one or two sentences, including the extremes it found.	Ch 9
REQ-1.5	Accuracy dashboard	The system re-runs the recorded benchmark on demand, computes its own accuracy and shows the result beside its answers.	Ch 10
REQ-1.6	Record and re-run an exchange	Every exchange is stored with the question, the derived request, the data snapshot identifier and the figure, and can be re-run later to reproduce the same figure.	Ch 11

The questionnaire responses support the six requirements above; no additional independent function was requested. Requirements arising later from the assigned customer or from real potential users will be added to this list and numbered in sequence.

Chapter 6

Functional Requirement REQ-1.1

6.1 Description

The system shall let a non-specialist ask a trajectory question in Chinese or English and see the structured request before anything is plotted.

6.2 System Input

Free text of 1–500 Unicode characters. The question may name an object, a place, a time range and a result type. Empty or over-length input is rejected.

6.3 Display

A question box, a waiting indicator and a read-only request panel listing the interpreted intent, object, time zone, area, chart type and any ambiguity.

6.4 System Processing

Parse the intent, object, time, place and chart fields; resolve relative times and place names; ask for confirmation when more than one reading is plausible; execute only after confirmation.

6.5 System Output

The confirmed request, with intent, time range, spatial scope, object filter, chart type, resolved place names, time zone and a confidence notice.

6.6 Other

Source `S1/S2` Q4, Q9 and Q13: both respondents often want to read a pattern they cannot extract themselves, both asked to be shown what the system understood and to confirm it before anything runs, and both rated support for questions asked in Chinese as very important. Feature F1 of the Task 1 proposal.

6.7 Constraints

The system shall not draw a result from an unconfirmed reading, and shall not modify the source records.

6.8 Data Handling

Store the original question, the normalised request, the resolution choices and the confirmation decision. Use UTF-8 and validate length.

6.9 Error Handling

If no interpretation is possible, list the missing fields and stop. If two readings conflict, ask the user. If the user rejects the reading, return to the question.

Chapter 7

Functional Requirement REQ-1.2

7.1 Description

The system shall retrieve and filter the trajectories named in the confirmed request, resolving place names against the records.

7.2 System Input

The confirmed request from REQ-1.1, a selected trajectory snapshot and the place-name index. Source data are read-only.

7.3 Display

A filter summary, progress and a sample table; time, area, object and count are listed in the order of the request.

7.4 System Processing

Apply the time, spatial and object filters, resolve place names to identifiers, and keep the snapshot and filter settings.

7.5 System Output

The matching trajectories plus count, time range, area, object count, snapshot identifier and any unresolved place names.

7.6 Other

Source S1/S2 Q3 and Q13: both respondents work with tabular data, and both rated showing where the data came from as important, so a result shall stay inspectable as a table with a stated source.

7.7 Constraints

The system shall not alter source records or silently substitute another area, time range or object set.

7.8 Data Handling

Use UTF-8, an explicit time zone and coordinate system; validate dates and coordinates and log the filters with the snapshot.

7.9 Error Handling

If nothing matches, state which filter failed. If a place or time is ambiguous, return to REQ-1.1. If data are unavailable, report it.

Chapter 8

Functional Requirement REQ-1.3

8.1 Description

The system shall choose a chart that fits the confirmed question and draw it without asking the user to choose a plotting tool.

8.2 System Input

The confirmed request, the matching trajectories, a chart category and a display size; the system supports map, line, bar, heat map and table.

8.3 Display

A titled chart with labelled axes or legend, time window, units, source label and export controls for PNG, PDF and Excel.

8.4 System Processing

Use maps for paths, lines for time trends, bars for comparisons and heat maps for density; explain any alternative chosen.

8.5 System Output

The rendered figure plus its chart specification, time zone, filters and a link to the matching trajectory set.

8.6 Other

Source S1/S2 Q7 and Q8: both respondents chose the heat map and the summary, and one of them also chose the map and the line chart; in the ranking the summary came first for both and the bar chart last or next to last. All five forms are therefore offered. Feature F3 of the Task 1 proposal.

8.7 Constraints

The system shall not truncate an axis without saying so, and shall not omit units, source, time window or data label.

8.8 Data Handling

Store the chart specification with the data snapshot; use Unicode labels and consistent time and coordinate formats.

8.9 Error Handling

If the data cannot support a chart, show the table and explain. If rendering fails, keep the data result and report the failure.

Chapter 9

Functional Requirement REQ-1.4

9.1 Description

The system shall describe the figure it has drawn in one or two plain-language sentences, including any pattern it can see and the largest and smallest values.

9.2 System Input

The rendered figure, the aggregates behind it, the confirmed request and the time, area and object context.

9.3 Display

The summary beside the figure, with units, time window and area stated, and any uncertainty marked.

9.4 System Processing

State the largest and smallest values, the trend, the concentration and the coverage; do not infer causes.

9.5 System Output

One or two sentences in the language of the question.

9.6 Other

Source `S1`/`S2` Q7 and Q8: both respondents named the plain-language summary as a wanted output and both ranked it first, ahead of every drawn form. Feature F4 of the Task 1 proposal.

9.7 Constraints

The system shall not state a cause it cannot show, and shall not present a low-confidence reading as certain.

9.8 Data Handling

Generate the summary from the values, never from the rendered image, so that the two cannot disagree.

9.9 Error Handling

If the data are too small or incomplete for a summary, say so rather than produce a misleading one.

Chapter 10

Functional Requirement REQ-1.5

10.1 Description

The system shall measure and display its own accuracy on the built-in trajectory benchmark, so that a user can judge how far to trust an answer.

10.2 System Input

The built-in benchmark of 9,608 question–trajectory pairs, the version of the system, and a request to run the benchmark now.

10.3 Display

The accuracy rate, the sample size, the version of the system and of the benchmark, the run time and the cases that failed.

10.4 System Processing

Run the same question-to-trajectory chain used in normal operation; count a case as correct only when the whole answer is correct.

10.5 System Output

Numerator, denominator, sample size, benchmark version, run timestamp and the list of failed cases.

10.6 Other

Source S1/S2 Q11, Q15 and Q17: both respondents asked to see the accuracy rate on the interface, and the errors they feared—a wrongly computed time, a misidentified place, a wrong value—are exactly the errors such a benchmark exposes. Feature F5 of the Task 1 proposal.

10.7 Constraints

The figure shall not be presented as the accuracy on the customer's own questions, and failed cases shall not be hidden.

10.8 Data Handling

Store the system and benchmark versions, the timestamp, the per-case result and the aggregate; keep the data local.

10.9 Error Handling

If the benchmark is missing or a run fails, report the accuracy as unavailable, or as the last successful run with its timestamp.

Chapter 11

Functional Requirement REQ-1.6

11.1 Description

The system shall store every exchange and let the user re-run it later, so that a figure can be checked after the fact.

11.2 System Input

The original question, the confirmed request, the data snapshot, the chart specification, the summary, the time zone, the system version and the timestamp.

11.3 Display

A history list, newest first, and a comparison view for a re-run.

11.4 System Processing

Treat the stored exchange as immutable; load its snapshot, regenerate the figure and compare the result with the stored one.

11.5 System Output

A re-runnable record, the regenerated figure and a pass or fail report on the comparison.

11.6 Other

Source `S1`/`S2` Q10 and Q18: both respondents want a result they can keep and check outside the system—a PDF in both cases, an image in the earlier format question, and a spreadsheet from one of them. Re-running a stored exchange is the team's own proposal (`S4`); the questionnaire did not ask for it. Feature F6 of the Task 1 proposal.

11.7 Constraints

The system shall not overwrite a stored exchange, and shall not silently re-run against a newer snapshot.

11.8 Data Handling

Store records as UTF-8; retain them for two years or 100,000 exchanges, whichever comes first.

11.9 Error Handling

If the snapshot is missing, report that the exchange cannot be reproduced and name the missing dependency.

Chapter 12

External Interface Requirement

12.1 Data Interfaces

The system reads recorded movement data from comma-separated (CSV) or Excel files on the local disk, and reads the benchmark from a file shipped with the installation. It writes nothing back to either. Column names and the refresh frequency shall be validated against a sample file supplied with the data; a file that fails validation is rejected with the failing columns named.

Results leave the system through an export rather than a write-back: a PNG image, a PDF or a spreadsheet file. Both respondents asked for a PDF and both had chosen an image in the earlier format question; one of them also asked for a spreadsheet (Q10, Q18).

12.2 User Interfaces

The system presents a single-page web interface in the user’s browser. The page carries a question box and panels for the interpreted request, the filters, the figure, the plain-language summary, the accuracy dashboard and the exchange history with its re-run comparison. A mobile layout is out of scope: questions are typed on a desktop or laptop with a pointing device. Both respondents worked that way, one on macOS and one on Windows (Q1, Q2), and neither asked for a phone or a tablet.

12.3 Other Interfaces

There are none. The system uses only the local file system, the browser and the local CPU and GPU, and opens no network connection of its own.

Chapter 13

Non-Functional Requirements

13.1 System Performance

Table 13-1: Performance requirements.

Req. Id	Description	Response time
PERF-1.1	Time from asking a question to receiving the figure and the summary, for a single question.	10–30 s; max 60 s
PERF-1.2	Time to narrow a result already on screen—adding a filter or changing the time window—and see the figure redrawn.	Under 10 seconds
PERF-1.3	Time to re-run one recorded exchange and rebuild the same figure from the record alone.	Under 30 seconds
PERF-1.4	Time to re-run the whole recorded benchmark and report a new accuracy figure; a batch job, not an interactive one.	Under 24 hours

PERF-1.1 is the project's headline commitment, carried over from Task 1 measure M1. Asked how long they would wait for one answer (Q14), one respondent accepted 30 s to one minute and the other 10–30 s; the target band takes the tighter answer and the 60 s maximum takes the looser one. The figures are measured on the reference laptop (24 GB of memory, 10 GB of data, one user, warm start). They were put to the respondents as a team proposal rather than set by the customer; Appendix A records this.

13.2 Information Security

Table 13-2: Information security requirements.

Req. Id	Description
SEC-1.1	Credentials and any personal data shall not be transmitted in plain text over any network.
SEC-1.2	Any personal data shall be held only in an environment that has passed full system testing; it shall not be held in a development or demonstration environment.
SEC-1.3	The system shall meet the information security policies of Macau University of Science and Technology and of the assigned customer. The customer has not named the policy documents; the team applies the University's general rules in the meantime.
SEC-1.4	The system shall not hold data that could identify an individual, and shall not transmit records to any third-party service without the customer's written agreement.

SEC-1.3 cites no document that the team has not read. Q19 records only that both respondents had heard of data-use rules; neither was asked to name one, and the assigned customer named none.

13.3 Availability and Retention

Table 13-3: Availability requirements.

Req. Id	Description
AVA-1.1	The system shall be available during the customer's working hours, 09:00–18:00, on working days.
AVA-1.2	Planned unavailability shall not exceed one working day per calendar month and shall be announced at least 24 hours in advance.
AVA-1.3	Unplanned unavailability shall not exceed four hours per calendar month.
AVA-1.4	A recorded exchange shall remain re-runnable for at least two years after it was made.

13.4 Capacity

Table 13-4: Capacity requirements.

Req. Id	Description	Value
CAP-1.1	Volume of recorded movement data the system must hold and query in year ONE.	10 GB
CAP-1.2	The same for year TWO.	30 GB
CAP-1.3	The same for year THREE.	60 GB
CAP-2.1	Concurrent users the system must serve in year ONE.	5
CAP-2.2	The same for year TWO.	10
CAP-2.3	The same for year THREE.	20
CAP-3.1	Number of exchanges that must remain stored and re-runnable.	100,000

The values form a planning envelope, not a commitment the customer set. The questionnaire does not ask about data volume or concurrent users, and the closing invitation to raise anything the questions had missed (Q20) produced no answer from either respondent, so the envelope rests on `S4` alone. Sources: `S3` and `S4`.

Chapter 14

Domain Requirements

The task description asks that the specification cover functional, non-functional *and* domain requirements. The supplied template has no section for the third kind, so they are recorded here.

Table 14-1: Requirements arising from the domain rather than from a user.

Req. Id	Description
DOM-1.1	Time shall be handled with an explicit time zone; a question naming a time of day shall be interpreted in the time zone of the records, not of the person asking.
DOM-1.2	Positions shall be handled in a stated coordinate system, and distances shall be computed on the sphere rather than on a flat map.
DOM-1.3	A place name in a question shall be resolved against the names as they appear in the records, and the resolution used shall be shown to the user.
DOM-1.4	Charts shall follow the conventions of the domain: a stated and labelled scale, a stated time window, and no truncation of an axis without saying so. *(Source: Task 1, §3.5 measure M4)*
DOM-1.5	The system shall comply with the Macau Personal Data Protection Act (Law No. 8/2005) and with the licence that governs the map data it uses. Both respondents had heard of rules covering privacy, mapping and cross-border transfer (Q19), but the questionnaire did not ask them to name a document, so the Act is taken from published law rather than from the customer; see Appendix A.

Appendix A

Source Register and Traceability

Part A of the task asks that the requirements be elicited from three kinds of source: the assigned customer, which is compulsory; real potential users; and members of the team. This appendix records which sources were actually used, what each one supplied, and where the record falls short.

A.1 Register of Elicitation Sources

Table A-1: Every source cited in this document, and how it was obtained.

Id	Source	Method	Date	Status
S0	Assigned customer: the last group on the Part A roster, which is Group 1’s customer under the roster rule	Contacted through the course group, more than once, and not reached	before 19 Oct 2026	Not obtained
S1	User representative R1, a member of another group in the class	Written questionnaire of 20 questions, returned as a platform export	19 Oct 2026, 11:36	Received
S2	User representative R2, a member of a further group in the class	The same instrument, returned as a platform export	20 Oct 2026, 09:22	Received
S3	The team’s Task 1 proposal	Written by the team	7 Oct 2026	Used
S4	Team planning assumptions about scale and availability	Written by the team	20 Oct 2026	Provisional
S5	Professional user: a researcher at ByteDance working on movement and footprint data	Reported to the team by Chen Hongzhou; no document supplied	20 Oct 2026	Informal

S5 is the weakest entry in the register and is marked as such. It records that a researcher at ByteDance who works on movement and footprint data described analysis of this kind as a live requirement in that organisation; it was passed on verbally and no document was supplied. Nothing in Chapters 2–14 is justified by S5 alone. It is recorded because it shows that the need is not confined to the two respondents, and because omitting it would make the account of what was gathered less complete than it is.

A.2 Status of the Assigned Customer

The roster rule makes the last team on the Part A roster the customer of the first team, so Group 1's assigned customer is the last group. That group was contacted through the course group more than once before 19 October 2026 and could not be reached.

Because the assigned customer could not be reached, the questionnaire was put instead to individual members of other groups in the class: R1 from one group and R2 from another. They answered as ordinary users of the system, not as a team and not as the customer.

The team has not presented a user representative as the assigned customer. Where a requirement rests on S0, the table in §A.4 says so and the requirement itself carries a visible marker. The obligation is therefore unmet rather than concealed.

A.3 Summary of the Questionnaire

One instrument was used for both respondents: a written questionnaire of twenty questions, sent through the course group and answered in the respondent's own time; the platform recorded the submission time shown in Table A-1. The first six questions establish the context of the respondent—operating system, device, the kind of data handled, whether they often want to read a pattern they cannot extract themselves, how they analyse data today, and which step they find hardest. Q7 and Q8 ask what form the answer should take, first by multiple choice and then by ranking. Q9 asks what the system should do when it has misread the question. Q10 and Q18 ask what the result must be exportable as. Q11 asks how much accuracy matters and Q13 rates five properties of the tool on a five-point scale. Q14 sets the acceptable waiting time and Q15 the errors the respondent fears. Q12 is a control question that instructs the respondent to pick a given option; both answered it correctly, so neither response was discarded as careless. Q16 invites a recent worked example, Q17 asks whether the accuracy rate should be shown, Q19 asks whether the respondent has heard of data-use rules, and Q20 is an open invitation to name anything the questions missed.

Two answers carry more weight than the rest. Q8 ranked the one-sentence summary first for both respondents, ahead of every drawn form, which is why the summary is a requirement of its own (Chapter 5) rather than a caption on a figure. Q9 was answered identically by both—show me what you understood and ask me to confirm—which is why REQ-1.1 stops for confirmation instead of plotting from its best guess.

Some questions produced nothing usable, and are recorded as such. Neither respondent could give a worked example when Q16 asked for one, and neither named an unforeseen constraint in Q20; none has been invented here. Q11 split, one respondent rating accuracy as very important and the other as neither important nor unimportant, but both then asked in Q17 to see the accuracy rate on the interface, and REQ-1.5 follows Q17. Neither respondent named a policy document, because Q19 did not ask for one.

A.4 Traceability Matrix

Table A-2: Each requirement against the source that supports it.

Requirement	Source	Support
REQ-1.1	S1, S2 (Q4, Q9, Q13); S3	Questions from people who cannot write a query; the reading shown and confirmed before it runs
REQ-1.2	S1, S2 (Q3, Q13); S3	Tabular data; the result stays inspectable as a table with a stated source
REQ-1.3	S1, S2 (Q7–Q8); S3	Heat map and summary chosen by both; map and line by one; all five forms ranked
REQ-1.4	S1, S2 (Q7, Q8); S3	The plain-language summary ranked first by both respondents
REQ-1.5	S1, S2 (Q11, Q15, Q17); S3	Accuracy requested on the interface; time, place and value named as the feared errors
REQ-1.6	S4; S1, S2 (Q10, Q18)	Team proposal; respondents want the result kept and checked outside the system
PERF-1.1	S4; S3 (measure M1); S1, S2 (Q14)	Waiting time asked for; band and ceiling set by the team
SEC-1.3	S0 (open); S1, S2 (Q19)	No policy document named; the University’s general rules apply
AVA-1.1–1.4	S4	Working hours and allowance proposed by the team
CAP-1.1–3.1	S4	Envelope proposed by the team; not put to the respondents
DOM-1.5	S0 (open); S1, S2 (Q19)	Statutory duty taken from published law, not from the customer

A.5 Confirmation and Limitations

The requirements in this document have not been confirmed by the assigned customer. They rest on S1, S2, S3 and S4, and five limits follow.

1. Part A source 1 is not satisfied. S0 is recorded as not obtained; the gap is stated here and in §1.1 rather than hidden.

2. The non-functional numbers are a team proposal, not values the customer set. Only the waiting time was put to the respondents in a form they could answer (Q14), and where the two answers differed the tighter one was taken as the target. The data volumes and the concurrent-user figures were never put to them, and Q20 produced no comment on them.

3. The institution’s software and licence policy was not available by name. SEC-1.3 records that the University’s general rules apply in the meantime, and no document is cited that the team has not read.

4. Both respondents answered as ordinary users rather than as transport or planning specialists, and neither could offer a worked example when asked for one (Q16). The domain requirements of Chapter 14 therefore rest on published practice rather than on an expert interview.

5. `S5` is informal. It shows that a professional user in industry wants this capability, but it came without a document, carries no figure and supports no individual requirement; no requirement in this document would fall if `S5` were withdrawn.

If the assigned customer replies before submission, S0 will be recorded as received, the requirements it touches will be re-checked against the reply, and this appendix revised before the report is handed in.

References

[1] Group 1, *TrajVis Studio: A Self-Service Platform for Natural-Language Trajectory Analysis*, Task 1 project proposal, Software Engineering, MUST, 2026.

[2] *User Requirements Specifications Description*, Software Engineering, MUST, 2026.09.29.

[3] *Task2-User Requirements Specifications Template*, Software Engineering, MUST, 2026.09.29.

[4] Project repository, https://github.com/chenhongzhou666/MUST-SE-TrajVis

[5] V. J. Wei et al., *Text-to-TrajVis*, Findings of ACL, 2026. https://aclanthology.org/2026.findings-acl.972/